A Practical Guide to GPT language model

TOPIC -- GPT LANGUAGE MODEL

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OpenAI, which created the first generative pre-trained transformer (GPT) in 2018, asserted in 2023 that "GPT" should be regarded as a brand of OpenAI. In April 2023, OpenAI revised the brand guidelines in its terms of service to indicate that other businesses using its API to run their AI services would no longer be able to include "GPT" in such names or branding. In May 2023, OpenAI engaged a brand management service to notify its API customers of this policy, although these notifications stopped short of making overt legal claims (such as allegations of trademark infringement or demands to cease and desist). As of November 2023, OpenAI still prohibits its API licensees from naming their own products with "GPT", but it has begun enabling its ChatGPT Plus subscribers to make "custom versions of ChatGPT" called GPTs on the OpenAI site. OpenAI's terms of service says that its subscribers may use "GPT" in the names of these, although it's "discouraged". Relatedly, OpenAI has applied to the United States Patent and Trademark Office (USPTO) to seek domestic trademark registration for the term "GPT" in the field of AI. OpenAI sought to expedite handling of its application, but the USPTO declined that request in April 2023. In May 2023, the USPTO responded to the application with a determination that "GPT" was both descriptive and generic. As of November 2023, OpenAI continued to pursue its argument. For any given type or scope of trademark protection in the U.S., OpenAI would need to establish that the term is actually "distinctive" to their specific offerings in addition to being a broader technical term for the kind of technology. Some media reports suggested in 2023 that OpenAI may be able to obtain trademark registration based indirectly on the fame of its GPT-based chatbot product, ChatGPT, for which OpenAI has separately sought protection (and which it has sought to enforce more strongly). Other reports have indicated that registration for the bare term "GPT" seems unlikely to be granted, as it is used frequently as a common term to refer simply to AI systems that involve generative pre-trained transformers. In any event, to whatever extent exclusive rights in the term may occur the U.S., others would need to avoid using it for similar products or services in ways likely to cause confusion. If such rights ever became broad enough to implicate other well-established uses in the field, the trademark doctrine of descriptive fair use could still continue non-brand-related usage. In the European Union, the European Union Intellectual Property Office registered "GPT" as a trade mark of OpenAI in spring 2023. However, since spring 2024 the registration is being challenged and is pending cancellation. In Switzerland, the Swiss Federal Institute of Intellectual Property registered "GPT" as a trade mark of OpenAI in spring 2023.

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Foundation models A foundation model is an AI model trained on broad data at scale such that it can be adapted to a wide range of downstream tasks. The most recent OpenAI's GPT-n series model is GPT-5. Other such models include Google's PaLM, a broad foundation model that has been compared to GPT-3 and has been made available to developers via an API, and Together's GPT-JT, which has been reported as the closest-performing open-source alternative to GPT-3 (and is derived from earlier open-source GPTs). Meta AI (formerly Facebook) also has a generative transformer-based foundational large language model, known as LLaMA. Foundational GPTs can also employ modalities other than text, for input and/or output. GPT-4 is a multi-modal LLM that is capable of processing text and image input (though its output is limited to text). Regarding multimodal output, some generative transformer-based models are used for text-to-image technologies such as diffusion and parallel decoding. Such kinds of models can serve as visual foundation models (VFMs) for developing downstream systems that can work with images.

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A foundational GPT model can be further adapted to produce more targeted systems directed to specific tasks and/or subject-matter domains. Methods for such adaptation can include additional fine-tuning (beyond that done for the foundation model) as well as certain forms of prompt engineering. An important example of this is fine-tuning models to follow instructions, which is of course a fairly broad task but more targeted than a foundation model. In January 2022, OpenAI introduced "InstructGPT" – a series of models which were fine-tuned to follow instructions using a combination of supervised training and reinforcement learning from human feedback (RLHF) on base GPT-3 language models. Advantages this had over the bare foundational models included higher accuracy, less negative/toxic sentiment, and generally better alignment with user needs. Hence, OpenAI began using this as the basis for its API service offerings. Other instruction-tuned models have been released by others, including a fully open version. Another (related) kind of task-specific models are chatbots, which engage in human-like conversation. In November 2022, OpenAI launched ChatGPT – an online chat interface powered by an instruction-tuned language model trained in a similar fashion to InstructGPT. They trained this model using RLHF, with human AI trainers providing conversations in which they played both the user and the AI, and mixed this new dialogue dataset with the InstructGPT dataset for a conversational format suitable for a chatbot. Other major chatbots currently include Microsoft's Bing Chat, which uses OpenAI's GPT-4 (as part of a broader close collaboration between OpenAI and Microsoft), and Google's competing chatbot Gemini (initially based on their LaMDA family of conversation-trained language models, with plans to switch to PaLM). Yet another kind of task that a GPT can be used for is the meta-task of generating its own instructions, like developing a series of prompts for 'itself' to be able to effectuate a more general goal given by a human user. This is known as an AI agent, and more specifically a recursive one because it uses results from its previous self-instructions to help it form its subsequent prompts; the first major example of this was Auto-GPT (which uses OpenAI's GPT models), and others have since been developed as well. Chain-of-thought reasoning is a prompting technique in which a language model generates intermediate reasoning steps before arriving at a final answer. This approach has been shown to improve performance on tasks requiring multi-step reasoning, such as mathematical problem solving and logical inference. By producing step-by-step explanations, the model is able to decompose complex problems into smaller, more manageable parts. This technique is often used in conjunction with few-shot prompting, where examples demonstrating the reasoning process are included in the input.